

Unit 1

Fundamentals of Computers

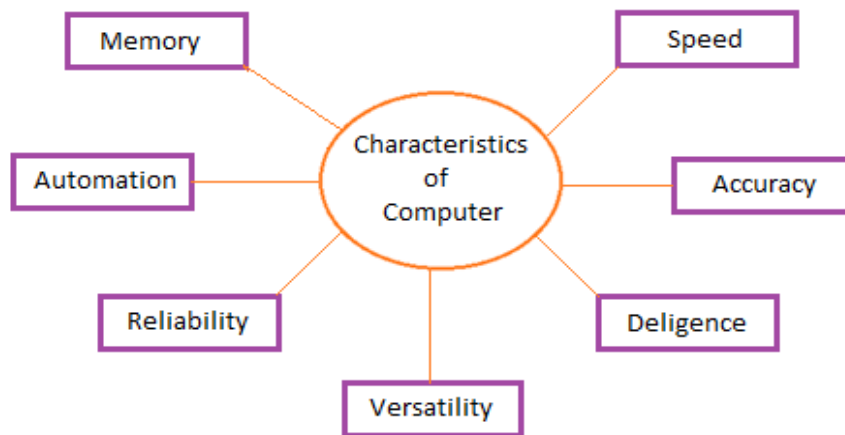
1.1 Introduction to Computers - Computer Definition:

- The word “Computer” comes from the /word “Compute” which means “to calculate”.
- Computer is a general purpose electronic device that is used to perform arithmetic and logical operations automatically.
- A computer consists of a central processing unit and some form of Memory.
- Computer is an electronic device which is capable of receiving information or data and perform a series of operations in accordance with a set of operations.
- This produces results in the form of data or information. Computer is a machine capable of solving problems and manipulating data. It accepts and processes the data by doing some mathematical and logical operations and gives us the desired output.
- Computer Full form:
 - C - Common
 - O - Operating
 - M - Machine
 - P - Purposely
 - U - Used for
 - T - Technological and
 - E - Educational
 - R – Research

1.2 Characteristics of Computer:

- **Speed**
 - A computer works with much higher speed and accuracy compared to humans while performing mathematical calculations. Computers can process millions (1,000,000) of instructions per second. The time taken by computers for their operations is microseconds and nanoseconds.
- **Accuracy**
 - Computers perform calculations with 100% accuracy. Errors may occur due to data inconsistency or inaccuracy.
- **Diligence**
 - A computer can perform millions of tasks or calculations with the same consistency and accuracy. It doesn't feel any fatigue or lack of concentration. Its memory also makes it superior to that of human beings.
- **Versatility**
 - Versatility refers to the capability of a computer to perform different kinds of works with same accuracy and efficiency.

- **Reliability**
 - A computer is reliable as it gives consistent result for similar set of data i.e., if we give same set of input any number of times, we will get the same result.
- **Automation**
 - Computer performs all the tasks automatically i.e. it performs tasks without manual intervention.
- **Memory**
 - A computer has built-in memory called primary memory where it stores data. Secondary storage are removable devices such as CDs, pen drives, etc., which are also used to store data.
- **NO IQ**
 - A Computer is not a magical device. It possesses no intelligence of its own. Its I.Q is zero, at least until today. It has to be told what to do and in what sequence.
- **No Feelings:**
 - Computers are devoid of emotions. They have no feelings and no instincts because they are machines.



1.3 Evolution and History of Computers:

Evolution:

- The first counting device was used by the primitive people.
- They used sticks, stones and bones as counting tools. As human mind and technology improved with time more computing devices were developed.
- one of the popular computing devices starting with the first to recent ones are described below;

Abacus

- The history of computer begins with the birth of abacus which is believed to be the first computer. It is said that Chinese invented Abacus around 4,000 years ago.
- It was a wooden rack which has metal rods with beads mounted on them. The beads were moved by the abacus operator according to some rules to perform arithmetic calculations. Abacus is still used in some countries like China, Russia and Japan.

Napier's Bones

- It was a manually-operated calculating device which was invented by John Napier (1550-1617) . In this calculating tool, he used 9 different ivory strips or bones marked with numbers to multiply and divide. So, the tool became known as "Napier's Bones. It was also the first machine to use the decimal point.

Pascaline

- Pascaline is also known as Arithmetic Machine or Adding Machine. It was invented between 1642 and 1644 by a French mathematician-philosopher Blaise Pascal. It is believed that it was the first mechanical and automatic calculator.
- Pascal invented this machine to help his father, a tax accountant. It could only perform addition and subtraction. It was a wooden box with a series of gears and wheels. When a wheel is rotated one revolution, it rotates the neighboring wheel. A series of windows is given on the top of the wheels to read the totals.

Stepped Reckoner or Leibnitz wheel

- It was developed by a German mathematician-philosopher Gottfried Wilhelm Leibnitz in 1673. He improved Pascal's invention to develop this machine. It was a digital mechanical calculator which was called the stepped reckoner as instead of gears it was made of fluted drums. See the following image;

Difference Engine

- In the early 1820s, it was designed by Charles Babbage who is known as "Father of Modern Computer". It was a mechanical computer which could perform simple calculations. It was a steam driven calculating machine designed to solve tables of numbers like logarithm tables.

Analytical Engine

- This calculating machine was also developed by Charles Babbage in 1830. It was a mechanical computer that used punch-cards as input. It was capable of solving any mathematical problem and storing information as a permanent memory.

Tabulating Machine

- It was invented in 1890, by Herman Hollerith, an American statistician. It was a mechanical tabulator based on punch cards. It could tabulate statistics and record or sort data or information. This machine was used in the 1890 U.S. Census. Hollerith also started the Hollerith's Tabulating Machine Company which later became International Business Machine (IBM) in 1924.

Differential Analyzer

- It was the first electronic computer introduced in the United States in 1930. It was an analog device invented by Vannevar Bush. This machine has vacuum tubes to switch electrical signals to perform calculations. It could do 25 calculations in few minutes.

Mark I

The next major changes in the history of computer began in 1937 when Howard Aiken planned to develop a machine that could perform calculations involving large numbers. In 1944, Mark I computer was built as a partnership between IBM and Harvard. It was the first programmable digital computer.

Generations of Computers

- A generation of computers refers to the specific improvements in computer technology with time. In 1946, electronic pathways called circuits were developed to perform the counting. It replaced the gears and other mechanical parts used for counting in previous computing machines.
- In each new generation, the circuits became smaller and more advanced than the previous generation circuits. The miniaturization helped increase the speed, memory and power of computers. There are five generations of computers which are described below

First Generation Computers

- The first generation (1946-1959) computers were slow, huge and expensive. In these computers, vacuum tubes were used as the basic components of CPU and memory. These computers were mainly depended on batch operating system and punch cards. Magnetic tape and paper tape were used as output and input devices in this generation;
- Some of the popular first generation computers are;
 - ENIAC (Electronic Numerical Integrator and Computer)
 - EDVAC (Electronic Discrete Variable Automatic Computer)
 - UNIVACI (Universal Automatic Computer)
 - IBM-701
 - IBM-650

Second Generation Computers

- The second generation (1959-1965) was the era of the transistor computers. These computers used transistors which were cheap, compact and consuming less power; it made transistor computers faster than the first generation computers.
- In this generation, magnetic cores were used as the primary memory and magnetic disc and tapes were used as the secondary storage. Assembly language and programming languages like COBOL and FORTRAN, and Batch processing and multiprogramming operating systems were used in these computers.
- Some of the popular second generation computers are;
 - IBM 1620
 - IBM 7094
 - CDC 1604
 - CDC 3600
 - UNIVAC 1108

Third Generation Computers

- The third generation computers used integrated circuits (ICs) instead of transistors. A single IC can pack huge number of transistors which increased the power of a computer and reduced the cost.

- The computers also became more reliable, efficient and smaller in size. These generation computers used remote processing, time-sharing, multi programming as operating system. Also, the high-level programming languages like FORTRAN-II TO IV, COBOL, PASCAL PL/1, ALGOL-68 were used in this generation.
- Some of the popular third generation computers are;
 - IBM-360 series
 - Honeywell-6000 series
 - PDP (Personal Data Processor)
 - IBM-370/168
 - TDC-316

Fourth Generation Computers

- The fourth generation (1971-1980) computers used very large scale integrated (VLSI) circuits; a chip containing millions of transistors and other circuit elements.
- These chips made these generation computers more compact, powerful, fast and affordable. These generation computers used real time, time sharing and distributed operating system. The programming languages like C, C++, DBASE were also used in this generation.
- Some of the popular fourth generation computers are;
 - DEC 10
 - STAR 1000
 - PDP 11
 - CRAY-1(Super Computer)
 - CRAY-X-MP (Super Computer)

Fifth Generation Computers

- In fifth generation (1980-till date) computers, the VLSI technology was replaced with ULSI (Ultra Large Scale Integration).
- It made possible the production of microprocessor chips with ten million electronic components.
- These generation computers used parallel processing hardware and AI (Artificial Intelligence) software. The programming languages used in this generation were C, C++, Java, .Net, etc.
- Some of the popular fifth generation computers are;
 - Desktop
 - Laptop
 - Note Book
 - Ultra Boo
 - Chrome book



Vacuum Tube



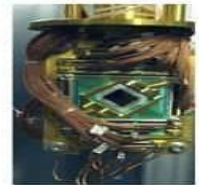
Transistors



Integrated Circuit



Microprocessor



Quantum Computer



1st Generation Computer



2nd Generation Computer



3rd Generation Computer

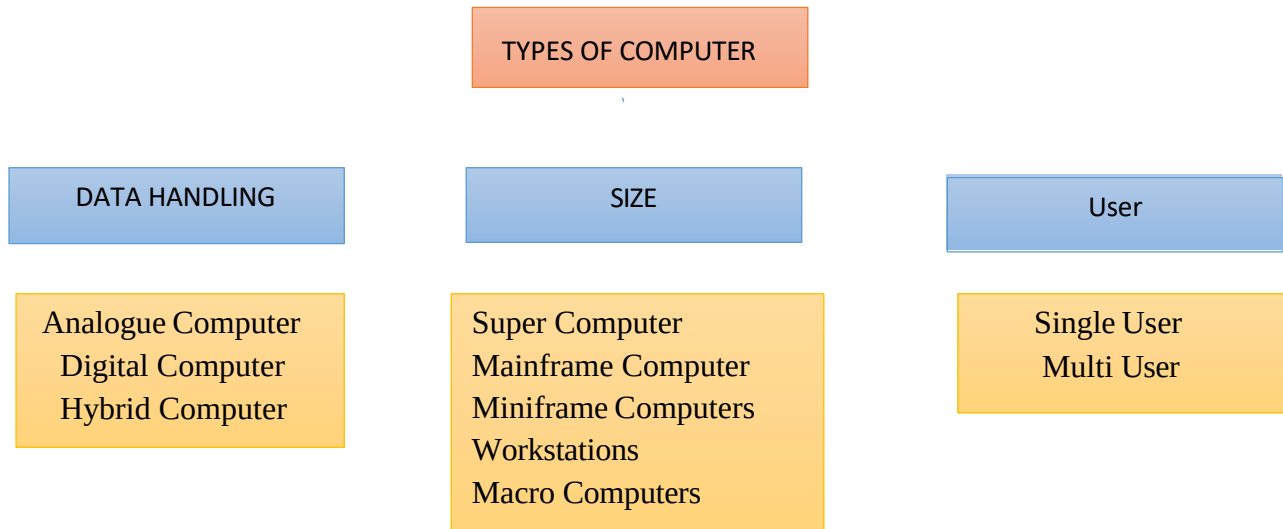


4th Generation Computer



5th Generation Computer

1.4 Types of computers:



On the basis of data handling capabilities, the computer is of three types:

- Analogue Computer
- Digital Computer
- Hybrid Computer

1) Analogue Computer

- Analogue computers are designed to process analogue data. Analogue data is continuous data that changes continuously and cannot have discrete values. We can say that analogue computers are used where we don't need exact values always such as speed, temperature, pressure and current.
- Analogue computers directly accept the data from the measuring device without first converting it into numbers and codes. They measure the continuous changes in physical quantity and generally render output as a reading on a dial or scale. Speedometer and mercury thermometer are examples of analogue computers.

Advantages of using analogue computers:

- It allows real-time operations and computation at the same time and continuous representation of all data within the range of the analogue machine.
- In some applications, it allows performing calculations without taking the help of transducers for converting the inputs or outputs to digital electronic form and vice versa.
- The programmer can scale the problem for the dynamic range of the analogue computer. It provides insight into the problem and helps understand the errors and their effects.

Types of analogue computers:

- **Slide Rules:** It is one of the simplest types of mechanical analogue computers. It was developed to perform basic mathematical calculations. It is made of two rods. To perform the calculation, the hashed rod is slid to line up with the markings on another rod.
- **Differential Analyzers:** It was developed to perform differential calculations. It performs integration using wheel-and-disc mechanisms to solve differential calculations.
- **Castle Clock:** It was invented by Al-Jarazi. It was able to save programming instructions. Its height was around 11 feet and it was provided with the display of time, the zodiac, and the solar and lunar orbits. This device also could allow users to set the length of the day as per the current season.

- **Electronic Analogue Computer:** In this type of analogue computer, electrical signals flow through capacitors and resistors to simulate physical phenomena. Here, the mechanical interaction of components does not take place. The voltage of the electrical signal generates the appropriate displays.

2) Digital Computer

- Digital computer is designed to perform calculations and logical operations at high speed. It accepts the raw data as input in the form of digits or binary numbers (0 and 1) and processes it with programs stored in its memory to produce the output. All modern computers like laptops, desktops including smartphones that we use at home or office are digital computers.

Advantages of digital computers:

- It allows you to store a large amount of information and to retrieve it easily whenever you need it.
- You can easily add new features to digital systems more easily.
- Different applications can be used in digital systems just by changing the program without making any changes in hardware
- The cost of hardware is less due to the advancement in the IC technology.
- It offers high speed as the data is processed digitally.
- It is highly reliable as it uses error correction codes.
- Reproducibility of results is higher as the output is not affected by noise, temperature, humidity, and other properties of its components.

3) Hybrid Computer

- Hybrid computer has features of both analogue and digital computers. It is fast like an analogue computer and has memory and accuracy like digital computers. It can process both continuous and discrete data. It accepts analogue signals and convert them into digital form before processing. So, it is widely used in specialized applications where both analogue and digital data is processed. For example, a processor is used in petrol pumps that converts the measurements of fuel flow into quantity and price. Similarly, they are used in airplanes, hospitals, and scientific applications.

Advantages of using hybrid computers:

- Its computing speed is very high due to the all-parallel configuration of the analogue subsystem.
- It produces precise and quick results that are more accurate and useful.
- It has the ability to solve and manage big equation in real-time.
- It helps in the on-line data processing.

On the basis of size, the computer can be of five types:

1) Supercomputer

- Supercomputers are the biggest and fastest computers. They are designed to process huge amount of data. A supercomputer can process trillions of instructions in a second. It has thousands of interconnected processors.
- Supercomputers are particularly used in scientific and engineering applications such as weather forecasting, scientific simulations and nuclear energy research. The first supercomputer was developed by Roger Cray in 1976.

Characteristics or applications of supercomputers:

- It has the ability to decrypt your password to enhance protection for security reasons.
- It produces excellent results in animations.
- It is used for virtual testing of nuclear weapons and critical medical tests.
- It can study and understand climate patterns and forecast weather conditions. It can run in NOAA's system (National Oceanic and Atmospheric Administration) that can execute any type of simple and logical data.
- It helps in designing the flight simulators for pilots at the beginner level for their training.
- It helps in extracting useful information from data storage centres or cloud system. For example, in insurance companies.
- It has played a vital role in managing the online currency world such as stock market and bitcoin.
- It helps in the diagnosis of various critical diseases and in producing accurate results in brain injuries, strokes, etc.
- It helps in scientific research areas by accurately analysing data obtained from exploring the solar system, satellites, and movement of Earth.
- It also used in a smog control system where it predicts the level of fog and other pollutants in the atmosphere.

2) Mainframe computer

- Mainframe computers are designed to support hundreds or thousands of users simultaneously. They can support multiple programs at the same time. It means they can execute different processes simultaneously. These features of mainframe computers make them ideal for big organizations like banking and telecom sectors, which need to manage and process high volume of data.
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Characteristics of Mainframe Computers:

- It can process huge amount of data, e.g. millions of transactions in a second in the banking sector.
- It has a very long life. It can run smoothly for up to 50 years after proper installation.
- It gives excellent performance with large scale memory management.
- It has the ability to share or distribute its workload among other processors and input/output terminals.
- There are fewer chances of error or bugs during processing in mainframe computers. If any error occurs, it can fix it quickly without affecting the performance.
- It has the ability to protect the stored data and other ongoing exchange of information and data.

Applications of mainframe computers:

- In health care, it enabled hospitals to maintain a record of their millions of patients in order to contact them for treatment or related to their appointment, medicine updates or disease updates.
- In the field of defence, it allows the defence departments to share a large amount of sensitive information with other branches of defence.

- In the field of education, it helps big universities to store, manage and retrieve data related to their courses, admissions, students, teachers, employees and affiliated schools and colleges.
- In the retail sector, the retail companies that have a huge customer base and branches use mainframe computers to handle and execute information related to their inventory management, customer management, and huge transactions in a short duration.

3) Miniframe or Minicomputer

- It is a midsize multiprocessing computer. It consists of two or more processors and can support 4 to 200 users at one time. Miniframe computers are used in institutes and departments for tasks such as billing, accounting and inventory management. A minicomputer lies between the mainframe and microcomputer as it is smaller than mainframe but larger than a microcomputer.

Characteristics of miniframe or minicomputer:

- It is light weight that makes it easy to carry and fit anywhere.
- It is less expensive than mainframe computers.
- It is very fast compared to its size.
- It remains charged for a long time.
- It does not require a controlled operational environment.

Applications of minicomputers:

A minicomputer is mainly used to perform three primary functions, which are as follows:

- Process control: It was used for process control in manufacturing. It mainly performs two primary functions that are collecting data and feedback. If any abnormality occurs in the process, it is detected by the minicomputer and necessary adjustments are made accordingly.
- Data management: It is an excellent device for small organizations to collect, store and share data. Local hospitals and hotels can use it to maintain the records of their patients and customers respectively.
- Communications Portal: It can also play the role of a communication device in larger systems by serving as a portal between a human operator and a central processor or computer.

4) Workstation

- Workstation is a single user computer that is designed for technical or scientific applications. It has a faster microprocessor, a large amount of RAM and high speed graphic adapters. It generally performs a specific job with great expertise; accordingly, they are of different types such as graphics workstation, music workstation and engineering design workstation.

Characteristics of workstation computer:

- It is a high-performance computer system designed for a single user for business or professional use.
- It has larger storage capacity, better graphics, and more powerful CPU than a personal computer.
- It can handle animation, data analysis, CAD, audio and video creation and editing.
- Any computer that has the following five features, can be termed as a workstation or can be used as a workstation.
- Multiple Processor Cores: It has more processor cores than simple laptops or computers.

- ECC RAM: It is provided with Error-correcting code memory that can fix memory errors before they affect the system's performance.
- RAID (Redundant Array of Independent Disks): It refers to multiple internal hard drives to store or process data. RAID can be of different types, for example, there can be multiple drives to process data or mirrored drives where if one drive does not work than other starts functioning.
- SSD: It is better than conventional hard-disk drives. It does not have moving parts, so the chances of physical failure are very less.
- Optimized, Higher end GPU: It reduces the load on CPU. E.g., CPU has to do less work while processing the screen output.

5) Microcomputer

- Microcomputer is also known as a personal computer. It is a general-purpose computer that is designed for individual use. It has a microprocessor as a central processing unit, memory, storage area, input unit and output unit. Laptops and desktop computers are examples of microcomputers. They are suitable for personal work that may be making an assignment, watching a movie, or at office for office work.

Characteristics of a microcomputer:

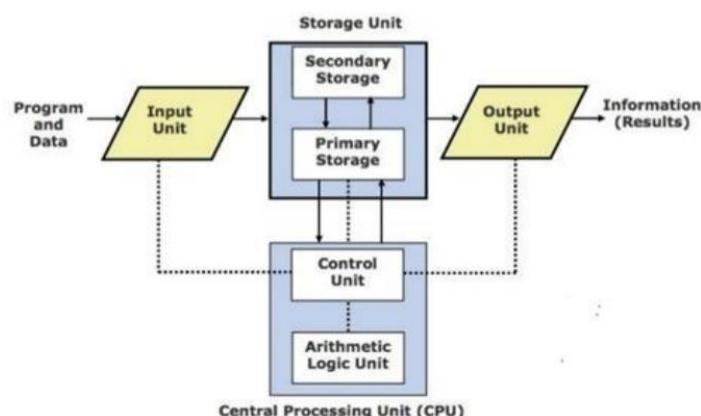
- It is the smallest in size among all types of computers.
- A limited number of software can be used.
- It is designed for personal work and applications. Only one user can work at a time.
- It is less expensive and easy to use.
- It does not require the user to have special skills or training to use it.
- Generally, comes with single semiconductor chip.
- It is capable of multitasking such as printing, scanning, browsing, watching videos, etc.

On the basis of User, the computer can be of two types:

Single User: A single user can access the computer at a particular time.

Multi User: Multi-user is a term that defines an operating system, computer program, or a game that allows use by more than one users of the same computer at the same time.

Basic organization of a computer setup:



Indicates flow of instructions and data

..... Indicates the control exercised by the control unit

- **Input Unit:** It accepts (or reads) instructions and data from outside. It converts these instructions and data in computer acceptable form. It supplies the converted instructions and data to the computer system for further processing. Central Processing Unit (CPU)
- **Control Processing Unit:** fetching instructions from memory, decoding the instructions, executing the instructions, performing arithmetic and logic operations, and storing the results back in memory.

Control unit(CU): control unit of a computer system manages and coordinates the operations of all other components of the computer system.

Arithmetic Logic Unit (ALU): Arithmetic logic unit of a computer system is the place, where the actual executions of instruction, takes place during processing operation.

- **Storage Unit:** Storage Units are responsible for storing all kinds of data and information.

Primary Storage Unit: It loses data on power dissipation(failures). It is used to hold running program instruction, data, intermediate results, and results of ongoing processing of jobs. **Secondary Storage Unit:** It is non-volatile (retains data even without power). It is used to hold stored program instructions and a large volume of information

- **Output Unit:** · It accepts the results produced by the CPU. It converts these coded results to human acceptable form. It supplies the converted results to outside world.

Difference between Primary memory and Secondary Memory

Primary storage	Secondary storage
Data is processed directly from storage.	Data must be moved to primary storage.
Provides faster access time.	Slow access time
Has slower storage capacity.	Has higher storage capacity.
Found inside the Central Processing Unit.	Found outside the Central Processing Unit.

Input / Output organizations:

Peripheral Devices

INPUT DEVICE: -

- The input device used to enter data and instruction in to the computer by the User.
- It Converts the given data into a form that is recognized by computer.
- Example: Keyboard, Mouse, OMR, OCR, MICR, Scanner, Web camera, Light pen.

1. KEYBOARD

- It is the most commonly used input device.
- It is used to enter the data into the computer.
- It consists of keys arrange similarly in a type writer.
- **Example:** Q W E R T Y U I O P ETC...

DIFFERENTS KINDS OF KEYS

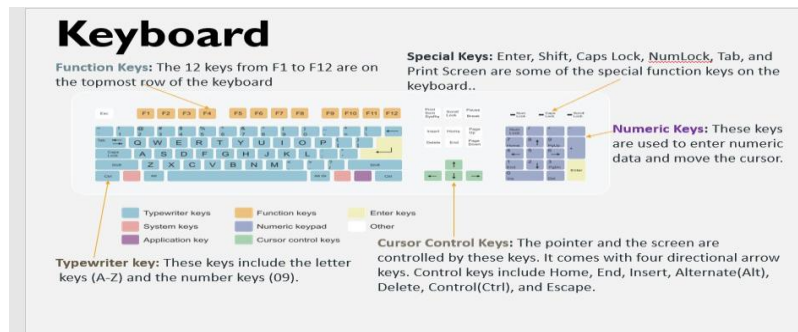
- Alphabet keys (A-Z)
- Numeric keys (0-9)
- Punctuation keys(.,?;:''=->,)
- Special keys(Alt,ctrl,Del,Tabs,..)

V. Functional Keys (F1 to F12)

EX: F5 – refresh computer

F2- save

- Number of keys in keyboard is 104 OR 108.
- Detects key pressed and generates ASCII Code and sent to computer.
[ASCII – American standard code for information interchange]



2. MOUSE:

- Mechanical operated user serial engine.
- It is a pointing device to controls movement of courser on screen.
- It is used to select objects and open apps in desktop.
- A ball **OR** IR [infrared rays] inside it.
- The movement of Ball converts into cursor on the screen.

Types of mouse:

there are mainly 3 types of mouse, they are:

1. Track Ball
2. Mechanical mouse
3. Optical mouse
4. Wireless mouse

a) TRACK BALL

- It is a type of a mouse
- A **trackball** is a [pointing device](#) consisting of a [ball](#) held by a socket containing sensors to detect a rotation of the ball.
- Users roll the ball to position the on-screen [pointer](#), using their [thumb](#), [finger](#), or the palm of the [hand](#), while using the fingertips to press the buttons.



b) MECHANICAL MOUSE

- It is a type of a mouse.
- It consisting of a small ball under its surface.
- The movement will be recognized as cursor on the screen by mechanical sensor.



c) OPTICAL MOUSE

- It is a type of mouse.
- It has a laser beam inside a mouse.
- The movement will be recognized as cursor on screen by this laser beam.



d) WIRELESS MOUSE

- It is a type of mouse.
- Wireless Mouse is one of the Laser mouse where you do not need to connect mouse wire from Laptop.
- In Wireless Mouse you need to battery to operate communication between USB receiver and Mouse.



3. OMR:

- It stands for Optical Mark reader.
- It reads marks made pencil or pen.
- **Application:** it is used to evaluate OMR sheets in competitive exams.
- **Working:** it focusses light on the OMR sheet after it will recognized and reflects light Back and marks will be read to the system.



4. OCR:

- It stands for optical character recognition or reader.
- It reads the printed characters includes the letter, number, special character
- **Application:** It used to scan a page front books.



5. MICR:

- It stands for magnetic Ink character recognition.
- It is used Recognize the Magnetic ink character.
- **Application:** it is used to process cheques in a banks.
- **Advantages of MICR:** - It can eliminate the manual errors and also can save the time and also can ensure accuracy and security.



6. SCANNER

- It is a input device.
- It is used to scan documents, images or photos.
- Scanning documents should be very clear otherwise it will have rejected.
- Scanned data will be converted into digital format.
- **Advantages:**
 - To avoid duplicate of data.
 - It reduces manual errors.
 - It can also save the time.
- **Application:** It is used in cyber center, to online transactions and to fill the online application.

Types of Scanner

1. Photo Scanner

A type of optical scanner that is mainly used to scan photographs. Photo scanners provide high resolution and color depth. They are smaller as compared to general-purpose scanners. Typically, a photo scanner has the ability to scan 3x5-inch or 4x6-inch photographs with higher resolution.

2. Flatbed scanner

Which scans documents on a flat surface.

These scanners do not require the document to be moved and are capable of capturing all of the document's elements. Flatbed scanners come in a couple of different sizes for standard paper and are effective for delicate materials, like documents that are fragile, including also vintage photographs and papers.

3. Handheld Scanner

The handheld scanner, on the other hand, is dragged over the page to be scanned. It scans physical documents into their digital forms, which makes it capable of storing, modifying, forward. and emailing digitally.

4. **Film Scanner**

A film scanner is a device that scans photographic film and transfers it to a computer. It scans without the need of any printmaking intermediates.

5. **Portable Scanner**

Portable scanners are designed in a way that can be easily carried around as they are small in size. Even some can be carried in the pockets, too, as they are as small as your PDAs. They are effective for text document scanning. They have limitations in terms of resolution. They are also available with a wireless facility.

OUTPUT DEVICE

- It is used to get the result after processing
- It presents information in human understandable form.
- **Example:** Monitor, Printer, Speaker, Projector, Plotter etc

1. **MONITOR**

- It is most commonly used output.
- It is used to display images, texts, graphics.
- The size of the monitor will be available 4:3 inches.
- The clarity of the monitor is depending on the number of pixels
- If pixel is more than the clarity is more.
- Most monitor have a 640 * 840, 600*400,1024*768
- The size of the monitor is 12,14,17,19, and 21 inches.

Types of monitor



CRT (Cathode Ray Tube)

Monitors are built very similarly to older (tube) television sets. They are heavy, bulky, take up a lot of desk space, and emit radiation.



LCD (Liquid Crystal Display) Monitors are thin and flat. They are light, compact, take up very little desk space emit no known radiation



LED(Light Emitting Diode)

a flat screen LCD display that uses light-emitting diodes (LEDs) as pixels to create visuals.

2. **PRINTER:**

- Output device used to print documents, Images and graphics on paper.
- It produces hard copy. Because the output will be in printed form and also in permanent form.
- **Applications:** It is used in banks, hospitals, Airports, Railway station, colleges, schools, etc...

Types of printers:

There are 2 types of printers.

- i. Impact printer
- ii. Non-impact printer

IMPACT PRINTERS	NON-IMPACT PRINTERS
➤ Printer which consisting of print head and the ink ribbon. ➤ Produces the output by striking the print head against ink ribbon placed between paper and print head. ➤ Prints multiple copies at a time. ➤ It's noisy ➤ It works slowly. ➤ It does not support transparency ➤ Speed is measured in number of characters per second. ➤ Example: Line printer, dot matrix printer.	➤ Printer doesn't consist print head and ink ribbon ➤ It doesn't have the striking mechanism. ➤ It doesn't support multiple copies. ➤ It's noiseless. ➤ It works faster ➤ It supports transparency ➤ Speed is measured in number of pages per minute. ➤ Example: Laser printer, inkjet printer, thermals printer.

Line Printer:

- Printer produces entire line at a time.
- Its speed is 150-3000 lines per minute.
- It's noisy
- Its quality is low.
- It can print only font style.
- It can't produce graphic image.
- **Application:** - It is used in stationaries.

Dot Matrix printer:

- Printer produces characters in the form of dots.
- Its speed is 300 cps[character per second.
- Example: :: :: ::
- It's quality is low.
- It is similar to type writer.
- It is slower.
- **Application:-**It is used in Medical stores, office, school, college.

Laser printer:

- printer uses laser beam to produce characters.
- Laser beam sensitized the printed area.
- This sensitized area attracts toner.
- This toner fused on paper.
- Its quality is high.
- It produces the graphic images.
- It also called as page printer.
- It's speed id 10-200 pages per minute.
- **Application:** - it is used in office, colleges, schools, browsing centers.

Ink jet printer:

- The printer produces character by spraying ink on the paper.
- Ink drop through the nozzle.
- Ink is fused through electrical charges.
- Finally, character is produced.
- It can take the printer is very high.
- It prints slower.
- Its speed is 100cps.
- **Applications:** it is used in stationaries, browsing centers, school, colleges, home.

Thermal printers:

- The printers produce characters by pushing electrically heated pins against paper.
- It is Inexpensive.
- The quality is low.
- It is darker when expose to sunlight and also fade up after a week.
- **Application:** fax machines, calculators, offices.

3. PLOTTER

- It is a type of output device.
- It used to draw the image using pens.
- It is used to get colour images.
- It is uses different colour pen.
- It produces images by drawing lines.
- It is costlier than the printer.
- **Application:** it is used in engineering fields, banners, flex.

A plotter is a device that prints high-quality graphics in a variety of color formats. It works in a similar way to a printer, although it has more advanced features. It is used to print large maps, architectural drawings, large-format printing, and create pictures, 3D postcards, advertising signs, charts, and various designs of the internal structure of building machines



4. SPEAKER:

It is a output device.

It produces voice output.

It used to listen to music and sounds.

Applications: Airlines, banks, Railway stations, home, telephone enquiry system.

MEMORY ORGANIZATIONS

MEMORY UNIT

- It is an essential component of computer.
- It is a storing device Used to stores Data, Programs, Information's, results.
- capacity is measured in bytes.

8bits =1 byte

1024 bytes=1kilo byte[KB]

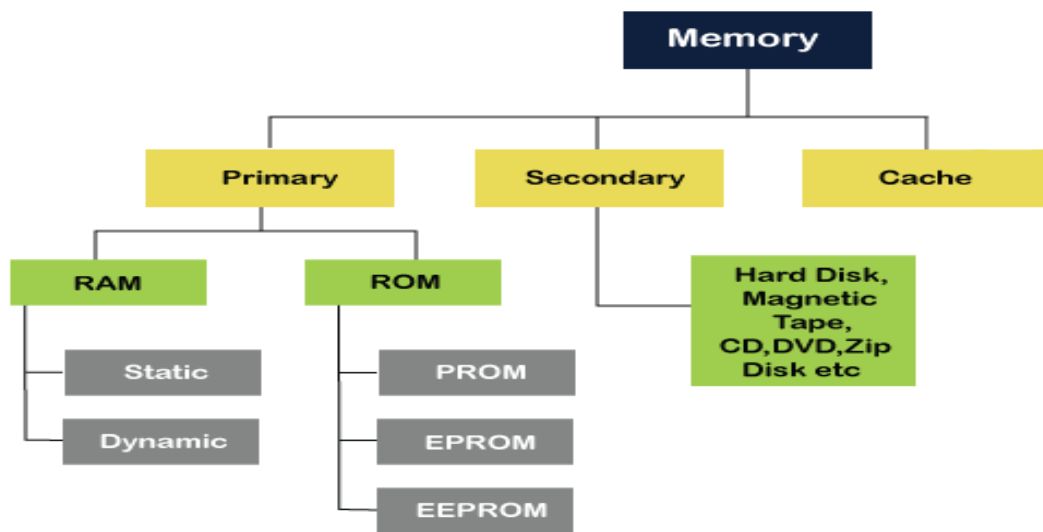
1024 KB = 1mega byte[MB]

1024MB=1 giga byte[GB]

1024GB=1tera byte

Bits stores in 0's and 1's in the memory.

TYPES OF MEMORY



2. PRIMARY MEMORY:

- It is also called as main memory.
- It stores Data, programs, Information that are currently needed by CPU.
- CPU id directly communicates with primary memory.
- It works in a faster way.

Primary memory functions.

It consists of a copy of system software (operating system).

It stores a copy of Application software

Ex : Ms office.

It stores the input data given through the Keyboard.

It also stores the result and these results is going to get after processing Output Device.

Types of primary memory

1. **RAM:** Random Access Memory
2. **ROM:** Read Only Memory

Difference between RAM and ROM

RAM	ROM
▪ Random Access Memory. ▪ Any memory Location can be access in a Random manner. ▪ We perform read and write operation. ▪ The storage capacity is high. ▪ It is expensive. ▪ It is faster. ▪ It is volatile. ▪ It loses its content when power goes off. ▪ It used for temprory information ▪ Types: i. Static RAM ii. Dynamic RAM	▪ Read Only Memory ▪ Random Property ▪ Perform only read operation. ▪ the storage capacity is low ▪ It is cheep ▪ It is slower. ▪ It is non-volatile. ▪ Retain content even power goes off. ▪ It is Used for store permanent information ▪ Types: i. PROM ii. EPROM iii. EEPROM

Types of RAM

Dynamic RAM	Static RAM
▪ It refresh content for every 2 mili seconds ▪ The storage capacity is high. ▪ It is expensive. ▪ It works in a faster way. ▪ It consume more electricity.	▪ Retain the information as long as power is off. ▪ The storage capacity is low. ▪ It is cheap. ▪ It works in a slower way. ▪ It consume Less electricity.

Types of ROM

1. PROM:

- Programmable Read only memory.
- We use PROM programmer for programming. User decides Contents of this memory.
- It stores data, Programs, Information permanently.
- User can write Data only once.
- User can't erase the content.
- Example printer of CD gives instructions to use the printer.

2. EPROM:

- Erasable programmable read only memory. Erase the contents of memory using UV rays.
- Procedure: It remove chip in computer.
- It exposes to UV rays for 20 minutes.
- It can't erase the contents of certain memory location. It is very cheap.
- It is widely used.
- It is reliable trust and depend upon.

3. EEPROM:

- Electrically erasable programming read only memory. Erase the contents of memory using electrical charges.
- Procedure:
 - No need to remove memory on computer.
 - Integrate circuit in the computer.

4. SECONDARY MEMORY

- Secondary memory is a **permanent storage** space to hold a large amount of data.
- Secondary memory is also known as external memory that representing the various storage media (hard drives, USB, CDs, flash drives and DVDs) on which the computer data and program can be saved on a long term basis.
 - ✗ it is cheaper and slower than the main memory.
- Unlike primary memory, secondary memory cannot be accessed directly by the CPU. Instead of that, secondary memory data is first loaded into the RAM (Random Access Memory) and then sent to the processor to read and update the data.
- Secondary memory devices also include magnetic disks like hard disk and floppy disks, an optical disk such as CDs and CDROMs, and magnetic tapes.

Features of Secondary Memory

- Its speed is slower than the primary / main memory.
- Store data cannot be lost due to non-volatile nature.
- It can store large collections of different types, such as audio, video, pictures, text, software, etc.
- All the stored data in a secondary memory cannot be lost because it is a permanent storage area, even the power is turned off.
- It has various optical and magnetic memories to store data.

Types of Secondary Memory

Hard Disk

- A hard disk is a computer's permanent storage device.
- It is a non-volatile disk that permanently stores data, programs, and files, and cannot lose store data when the computer's power source is switched off.
- It is a large storage device, found on every computer or laptop for permanently storing installed software, music, text documentation, videos, operating system, and data until the user did not delete.

Floppy Disk

- A floppy disk is a secondary storage system that consisting of thin, flexible magnetic coating disks for holding electronic data such as computer files.
- It is also known as Floppy Diskette that comes in three sizes like 8 inches, 5.5 inches and 3.5 inches.
- The stored data of a floppy disk can be accessed through the floppy disk drive

CD (Compact Disc)

- A CD is an optical disk storage device, stands for Compact Disc.
- It is a storage device used to store various data types like audio, videos, files, OS, Back-Up file, and any other information useful to a computer.

- The CD has a width of 1.2 mm and 12 cm in height, which can store approximately 783 MB of data size

Types of CDs

1. **CD-ROM (Compact Disc Read Only Memory):** It is mainly used for bulk size mass like audio CDs, software and computer games at the time of manufacture. Users can only read data, text, music, videos from the disc, but they cannot modify or burnt it.
2. **CD-R (Compact Disc Recordable):** The type of Compact Disc used to write once by the user; after that, it cannot be modified or erased.
3. **CD-RW (Compact Disc Rewritable):** It is a rewritable CD disc, often used to write or delete the stored data.

DVD Drive/Disc

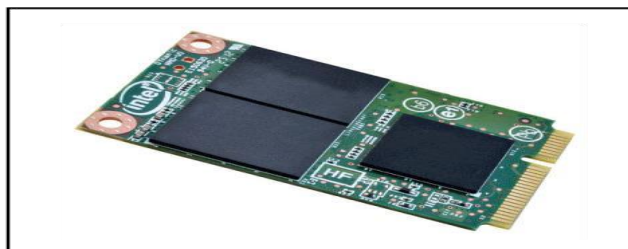
- DVD is an optical disc storage device, stands for **Digital Video Display or Digital Versatile Disc**.
- It has the same size as a CD but can store a larger amount of data than a compact disc.
- It was developed in **1995** by Sony, Panasonic, Toshiba and Philips four electronics companies.
- DVD drives are divided into three types, such as
DVD ROM (Read Only Memory),
DVD R (Recordable)
DVD RW (Rewritable or Erasable).
- It can store multiple data formats like audio, videos, images, software, operating system, etc.
- The storing capacity of data in DVD is 4.7 GB to 17 GB.

Pen Drive

- A pen drive is a portable device used to permanently store data and is also known as a USB flash drive.
- It is commonly used to store and transfer the data connected to a computer using a USB port.
- it uses an integrated circuit chip that stores the data.
- It allows the users to store and transfer data like audio, videos, images, etc from one computer to any USB pen drive.
- The storing capacity of pen drives from 64 MB to 128 GB or more.

5. CACHE MEMORY

- It is a small-sized chip-based computer memory that lies between the CPU and the main memory.
- It is a faster, high performance and temporary memory to enhance the performance of the CPU.
- It stores all the data and instructions that are often used by computer CPUs.
- It also reduces the access time of data from the main memory.
- It is faster than the main memory, and sometimes, it is also called CPU memory because it is very close to the CPU chip.



TYPES OF SOFTWARE

- The term “software” refers to the set of computer programs, procedures, and associated documents which describe the programs and how they are to be used.
- Relationship between hardware and software are described as follows: Both are necessary to the useful jobs and both complement each other
- The same hardware can be loaded to different software to make a computer system do different jobs.

Except for upgrades hardware is normally a one-time expense whereas software is a continuing expense. **Types of software include:**

I. System software

It is a set of one or more programs designed to control the operation and extending the processing capability of a computer system. The computer's system software performs these functions:

- Supports development of other application software
- Supports execution of other application software
- Monitors the effective use of various hardware resources, such as CPU, memory etc.
- Communicates with and monitors the operation of peripheral devices such as printers, disk, tape etc...

It also helps hardware components work together and provides support for development and execution of application software.

Some most commonly used system software are:

1. Operating system: every computer has an operating system software which takes care of effective and efficient utilization of all hardware and software components of the computer.

2. Programming language translator: They are the system software which transform information prepared by the programmers in programming language into a form that can be interpreted and executed by a computer system.

3. Device Driver: A device driver is a program or software that controls a device and helps that device to perform its functions. Every device like a printer, mouse, modem, etc. needs a driver to connect with the computer system externally. So, when you connect a new device with your computer system, first you need to install the driver of that device so that your operating system knows how to control or manage that device.

II. Application Software

Software that performs special functions or provides functions that are much more than the basic operation of the computer is known as application software. Or in other words, application software is designed to perform a specific task for end-users. It is a product or a program that is designed only to fulfill end-users' requirements. It includes word processors, spreadsheets, database management, inventory, payroll programs, etc.

There are different types of application software and those are:

- 1. General Purpose Software:** This type of application software is used for a variety of tasks and it is not limited to performing a specific task only. For example, MS-Word, MS-Excel, PowerPoint, etc.
- 2. Customized Software:** This type of application software is used or designed to perform specific tasks or functions or designed for specific organizations. For example, railway reservation system, airline reservation system, invoice management system, etc.

III. Utility software

Utility software is software designed to help analyze, configure, optimize or maintain a computer. It is used to support the computer infrastructure - in contrast to application software, which is aimed at directly performing tasks that benefit ordinary users. The computers utility software performs these functions:

Anti-virus utilities scan for computer viruses and block or remove them

- Backup software makes copies of all information stored on a disk and restores either the entire disk (aka Disk cloning) in an event of disk failure or selected files that are accidentally deleted or corrupted. Undeleting utilities are sometimes more convenient.
- Cryptographic utilities encrypt and decrypt streams and files.
- Data compression utilities output a shorter stream or a smaller file when provided with a stream or file.
- HTML checkers validate HTML code and check links
- Debuggers typically permit the examination and modification of data and program instructions in memory and on disk.
- Disk checkers scan an operating hard drive and check for logical (file system) or physical error
- Sort/Merge programs arrange records (lines) of a file into a specified sequence

COMPUTER LANGUAGES

A computer language is used by a programmer to instruct a computer what he or she wants Computer languages can be classified into three types:

- 1. Machine language**
- 2. Assembly language**
- 3. High level language**

1. Machine language

Although computers can be programmed many different languages, there is only one language that is understood by the computer without using a translating program that is machine language. A machine language instruction has a two-part form

The first part of instruction is the operation code that tells the computer what function to perform and the second part is the operand which tells the computer where to find or store data which are to be manipulated. Every computer has a set of operation codes called the instruction set. Each operation in the instruction set performs a specific basic operation or function.

Advantages:

- a. Programs can be executed very fast by the computer (No translation of the program is required)

Drawbacks:

- a. Machine dependent
- b. Error prone: the programmer must remember the opcodes and must keep track of storage locations
- c. Difficult to modify: Checking machine instructions to locate errors is very difficult and time consuming.

2. Assembly language

Assembly language was introduced in 1952 helped in overcoming the above listed limitations of machine language programming.

A language which allows instructions and storage locations to be represented by letters and symbols instead of numbers is called an assembly language.

Advantages of assembly language:

- Easier to understand and use
- Easier to locate errors and to modify them
- Easily re-locatable

Drawbacks:

- Machine dependent
- Knowledge of hardware is required
- Machine level coding

3. High level language

Machine and assembly languages have limitations as mentioned above so they are referred to as low level languages. High level languages were designed to overcome the limitations of low level programming languages.

This is the language which is understandable by the Users.

It consists English Language to write code.

Characteristics of high level languages are:

- a. They are machine independent. The program written in a high level language can be ported and executed on any computer which has the software for high level language translator.
- b. They do not require programmers to know anything about the internal structure of computer.
- c. They don't deal with machine level coding. They deal with high level coding, enabling programmers to write instructions using English words and familiar mathematical symbols and expressions.

The advent of high level languages has enabled the use of computers to solve the problems even by non-expert users, which has in turn resulted in creation of large number of computer applications in diverse areas, leading to use of computers in every occupation.

Advantages:

1. Machine independence
2. Easier to learn and use
3. Fewer errors
4. Low program preparation cost
5. Better documentation and easier to maintain

Drawbacks:

1. Lower efficiency: Programs written in high level languages take more time to execute and require more main memory space.
2. Less flexibility: They are less flexible than assembly level programs because they do not have instructions or mechanisms to control the computer's CPU, memory and registers.

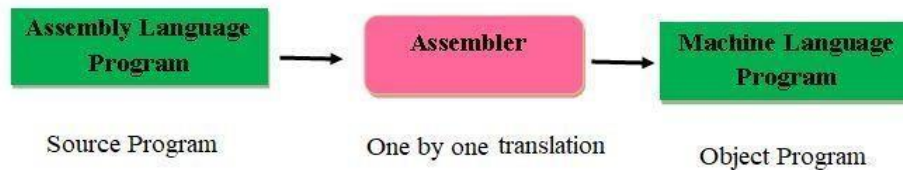
TRANSLATOR PROGRAMS:

1. Assembler:

A computer can directly execute machine level programs which use numbers for representing instructions. Hence assembly level programs must be translated to machine level before executing it. This translating job is done by an assembler.

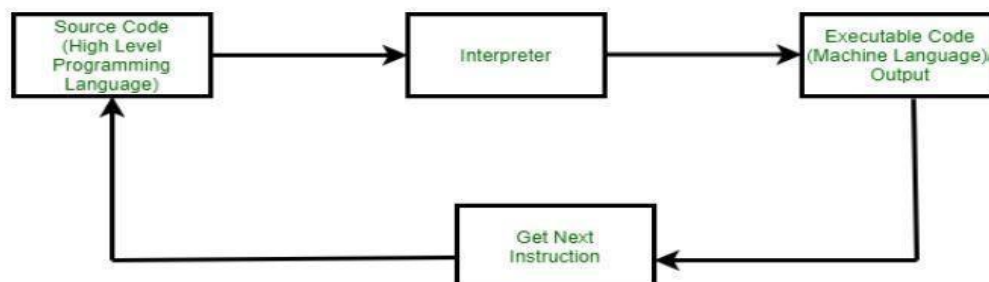
It is so called because in addition to translating an assembly level program to machine level, it also “assembles” the machine language programs in the main memory of the computer, making it ready for execution.

How Assembler Works



2. Interpreter:

It is another type of translator that is used for translating programs written in high level languages. It takes one statement of high level language program translates it into machine language instructions and immediately executes the resulting machine language instructions. In case of an interpreter the translation and execution are alternate for each statement encountered in high level languages. It differs from a compiler which merely translates entire source program into an object program and is not involved in execution.



3. Compiler:

The high level language programs must be converted into its equivalent machine language program before it can be executed. This is done with help of compiler. A compiler is more sophisticated than an assembler. It is so called because it compiles a set of machine language instructions for every program instruction of high level language.

Since high level language instructions are macro instructions, the compiler translates each high level language instruction into a set of machine language instruction rather than a single machine language instruction.

